CBSE Class 12 Chemistry — Chemical Kinetics

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

Q1. The slope in the plot of $[R]$ vs. time for a zero-order reaction is: (A) $\frac{+k}{2.303}$ (B) $-k$ (C) $\frac{-k}{2.303}$ (D) $+k$ [PYQ 2023] [Confidence: Very High]

Q2. Which of the following statements is true? (a) Molecularity of reaction can be zero or a fraction. (b) Molecularity has no meaning for complex reactions. (c) Molecularity of a reaction is an experimental quantity. (d) Reactions with the molecularity three are very rare but are fast. [PYQ SQP 2023] [Confidence: Very High]

Q3. For a reaction $2A \rightarrow 3B$, the rate of reaction is equal to: (A) $\frac{+3}{2}\frac{d[B]}{dt}$ (B) $\frac{+2}{3}\frac{d[B]}{dt}$ (C) $\frac{+1}{3}\frac{d[B]}{dt}$ (D) $+\frac{2d[B]}{dt}$ [PYQ 2023] [Confidence: High]

Q4. If the initial concentration of substance A is $1.5\ M$ and after 120 seconds the concentration of substance A is $0.75\ M$, the rate constant for the reaction if it follows zero-order kinetics is: (a) $0.00625\ mol\ L^{-1}\ s^{-1}$ (b) $0.00625\ s^{-1}$ (c) $0.00578\ mol\ L^{-1}\ s^{-1}$ (d) $0.00578\ s^{-1}$ [PYQ SQP 2023] [Confidence: High]

Q5. For the reaction, $A + 2B \rightarrow AB_2$, the order w.r.t. reactant A is 2 and w.r.t. reactant B is 1. What will be the change in rate of reaction if the concentration of A is doubled and B is halved? (a) increases four times (b) decreases four times (c) increases two times (d) no change [PYQ SQP 2022] [Confidence: Very High]

Q6. Arrhenius equation can be represented graphically by plotting $\ln k$ vs $\frac{1}{T}$. The (i) intercept and (ii) slope of the graph are: (a) (i) $\ln A$ (ii) $\frac{E_a}{R}$ (b) (i) A (ii) E_a (c) (i) $\ln A$ (ii) $\frac{-E_a}{R}$ (d) (i) A (ii) $-E_a$ [PYQ SQP 2022] [Confidence: Very High]

Q7. Which radioactive isotope would have the longer half-life, ^{15}O or ^{19}O? (Given rate constants for ^{15}O and ^{19}O are $5.63 \times 10^{-3}\ s^{-1}$ and $k = 2.38 \times 10^{-2}\ s^{-1}$ respectively.) (a) ^{15}O (b) ^{19}O (c) Both will have the same half-life (d) None of the above [PYQ SQP 2022] [Confidence: Medium]

Q8. Assertion (A): Rate of reaction is a measure of change in concentration of reactant with respect to time. Reason (R): Rate of reaction is a measure of change in concentration of product with respect to time. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ Passage Based] [Confidence: High]

Q9. Assertion (A): For a reaction: $P + 2Q \rightarrow Products$, Rate = $k[P]^{1/2}[Q]^1$, so the order of reaction is 1.5. Reason (R): Order of reaction is the sum of stoichiometric coefficients of the reactants. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ Passage Based] [Confidence: Very High]

Q10. Assertion (A): The unit of k is independent of the order of reaction. Reason (R): The unit of k is $moles\ L^{-1}\ s^{-1}$. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ Passage Based] [Confidence: High]

Q11. Assertion (A): The molecularity of a complex, multi-step reaction has no meaning. Reason (R): A complex reaction takes place in several elementary steps, and molecularity is only defined for individual elementary steps. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [Practice] [Confidence: Very High]

Q12. Identify the order of reaction from the following unit for its rate constant: $L\ mol^{-1}\ s^{-1}$. [PYQ Term-2 2022] [Confidence: Very High]

Q13. Write the slope value obtained in the plot of $\ln[R]$ vs time for a first order reaction. [PYQ 2020] [Confidence: Very High]

Q14. For a reaction: $H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$; Rate = k. Write the order and molecularity of this reaction. [PYQ 2016] [Confidence: High]

Q15. The conversion of molecules A to B follows second order kinetics. If concentration of A is increased to three times, how will it affect the rate of formation of B? [PYQ 2023] [Confidence: Very High]

Q16. Write the expression of the integrated rate equation for a zero-order reaction. [PYQ Term-2 2022] [Confidence: High]

Q17. Define a Pseudo first order reaction. [PYQ 2023] [Confidence: Very High]

Q18. What is the effect of adding a catalyst on the activation energy (E_a) of a reaction? [PYQ 2017] [Confidence: Very High]

Q19. What is the effect of adding a catalyst on the Gibbs free energy (ΔG) of a reaction? [PYQ 2017] [Confidence: Very High]

Q20. The half-life of a first order reaction is $10\ minutes$. What is the value of its rate constant k? [Practice] [Confidence: Medium]

2-Mark Questions (Short Answer)

Q21. Radioactive decay follows first-order kinetics. The initial amount of two radioactive elements X and Y is $1\ gm$ each. What will be the ratio of X and Y after two days if their half-lives are $12\ hours$ and $16\ hours$ respectively? [PYQ SQP 2023] [Confidence: High]

Q22. The hypothetical reaction $P + Q \rightarrow R$ is half order w.r.t 'P' and zero order w.r.t 'Q'. What is the unit of rate constant for this reaction? [PYQ SQP 2023] [Confidence: High]

Q23. How will the rate of a chemical reaction be affected when: (a) the surface area of the reactant is reduced, and (b) the temperature of the reaction is increased? [PYQ 2020]

[Confidence: Very High]

Q24. In the given reaction $A + 3B \rightarrow 2C$, the rate of formation of C is $2.5 \times 10^{-4}\ mol\ L^{-1}\ s^{-1}$. Calculate the rate of reaction and the rate of disappearance of B. [PYQ 2020] [Confidence: Very High]

Q25. For the reaction $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$, the rate of formation of $NO_2(g)$ is $2.8 \times 10^{-3}\ M\ s^{-1}$. Calculate the rate of disappearance of $N_2O_5(g)$. [PYQ 2018] [Confidence: Very High]

Q26. Write units of rate constant for zero order and for second order reactions if the concentration is expressed in $mol\ L^{-1}$ and time in seconds. [PYQ 2015] [Confidence: Very High]

Q27. For a reaction: $2H_2O_2 \xrightarrow{I^-/alkaline} 2H_2O + O_2$, the proposed mechanism is: (1) $H_2O_2 + I^- \rightarrow H_2O + IO^-$ (slow) (2) $H_2O_2 + IO^- \rightarrow H_2O + I^- + O_2$ (fast) (A) Write the rate law for the reaction. (B) Write the overall order of reaction. [PYQ 2019] [Confidence: High]

Q28. Draw the plot of $\ln k$ vs $\frac{1}{T}$ for a chemical reaction. What does the intercept represent? What is the relation between the slope and E_a? [PYQ 2019] [Confidence: Very High]

Q29. A first order reaction takes $20\ minutes$ for 50% completion. Calculate the time required for 90% completion of this reaction. ($\log 2 = 0.3010$) [PYQ Term-2 2022] [Confidence: Very High]

Q30. The decomposition of NH_3 on a platinum surface is a zero order reaction. If the rate constant (k) is $4 \times 10^{-3}\ M\ s^{-1}$, how long will it take to reduce the initial concentration of NH_3 from $0.1\ M$ to $0.064\ M$? [PYQ 2019] [Confidence: High]

Q31. Define half-life of a reaction. Write the expression of half-life for a zero order reaction and a first order reaction. [PYQ 2014] [Confidence: Very High]

Q32. Explain how and why the rate of reaction for a given reaction will be affected when the temperature at which the reaction was taking place is decreased. [PYQ SQP 2022] [Confidence: High]

3-Mark Questions (Short Answer)

Q33. A first-order reaction is 40% completed in $80\ minutes$. Calculate the value of the rate constant (k). In what time will the reaction be 90% completed? (Given: $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 6 = 0.7782$) [PYQ 2020] [Confidence: Very High]

Q34. The rate of a reaction triples when the temperature changes from $298\ K$ to $318\ K$. Calculate the energy of activation of the reaction assuming that it does not change with temperature. (Given $R = 8.314\ J\ K^{-1}\ mol^{-1}$, $\log 3 = 0.4771$) [PYQ 2025-26 SQP] [Confidence: Very High]

Q35. The rate constants of a reaction at $200\ K$ and $500\ K$ are $0.02\ s^{-1}$ and $0.20\ s^{-1}$ respectively. Calculate the value of E_a. (Given $2.303R = 19.15\ J\ K^{-1}\ mol^{-1}$, $\log 10 = 1$)

[PYQ SQP 2023] [Confidence: High]

Q36. A first order reaction is 50% complete in $30\ minutes$ at $300\ K$ and in $100\ minutes$ at $320\ K$. Calculate the activation energy (E_a) for the reaction. ($R = 8.314\ J\ K^{-1}\ mol^{-1}$) [PYQ 2023] [Confidence: Very High]

Q37. A first order reaction is 50% completed in $40\ minutes$ at $300\ K$ and in $20\ minutes$ at $320\ K$. Calculate the activation energy of the reaction. (Given: $\log 2 = 0.3010$, $\log 4 = 0.6021$, $R = 8.314\ J\ K^{-1}\ mol^{-1}$) [PYQ 2018] [Confidence: High]

Q38. The rate of a reaction becomes four times when the temperature changes from $293\ K$ to $313\ K$. Calculate the energy of activation (E_a) of the reaction assuming that it does not change with temperature. ($R = 8.314\ J\ K^{-1}\ mol^{-1}$, $\log 4 = 0.6021$) [PYQ 2019] [Confidence: Very High]

Q39. The rate of a first order reaction increases from 2×10^{-2} to 4×10^{-2} when the temperature changes from $300\ K$ to $310\ K$. Calculate the activation energy (E_a). ($\log 2 = 0.3010$, $R = 8.314\ J\ K^{-1}\ mol^{-1}$) [PYQ 2015] [Confidence: High]

Q40. For a reaction $2NH_3(g) \xrightarrow{Pt} N_2(g) + 3H_2(g)$; $Rate = k$. Write the order and molecularity of this reaction. Also, write the unit of the rate constant k. Why does a bimolecular reaction sometimes become a zero-order reaction? [PYQ 2016 / 2014] [Confidence: Very High]

Q41. The rate constant for the first order decomposition of N_2O_3 is given by the following equation: $k = (2.5 \times 10^{14}\ s^{-1})e^{-25000/T}$. Calculate E_a for this reaction and the rate constant if its half-life period is $300\ minutes$. [PYQ 2020] [Confidence: Very High]

Q42. Show mathematically that for a first order reaction, the time required for 99% completion is exactly twice the time required for 90% completion. [Practice] [Confidence: High]

5-Mark Questions (Long Answer / Case-Study)

Q43. Case Study: Energy Profile and Catalysis Consider the graph for the reaction $H_2 + I_2 \rightleftharpoons 2HI$. The y-axis represents Potential Energy (kJ/mol) and x-axis represents Reaction Coordinate. The energy of reactants is $110\ kJ/mol$, the activated complex is at $140\ kJ/mol$, and products are at $100\ kJ/mol$. (A) Calculate the enthalpy of reaction (ΔH). (1 Mark) (B) Calculate the activation energy for the backward reaction. (1 Mark) (C) How will the presence of a positive catalyst affect the rate of this reaction and its activation energy? (2 Marks) (D) Does the catalyst change the equilibrium constant of the reaction? Explain. (1 Mark) [PYQ 2025-26 SQP] [Confidence: Very High]

Q44. (A) Derive the integrated rate equation for a first-order reaction: $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$. (B) A first-order reaction has a specific rate constant of $1.15 \times 10^{-3}\ s^{-1}$. How long will $5\ g$ of this reactant take to reduce to exactly $3\ g$? (Given: $\log 5 = 0.6990$, $\log 3 = 0.4771$) [PYQ / Practice] [Confidence: Very High]

Q45. For a reaction, $A + 2B \rightarrow 2C + D$, the following data was obtained experimentally: Exp 1: $[A]_0 = 0.1\ M$, $[B]_0 = 0.1\ M$, Initial Rate = $2.0 \times 10^{-3}\ M\ s^{-1}$ Exp 2: $[A]_0 = 0.2\ M$, $[B]_0 = 0.1\ M$, Initial Rate = $4.0 \times 10^{-3}\ M\ s^{-1}$ Exp 3: $[A]_0 = 0.2\ M$, $[B]_0 = 0.2\ M$, Initial Rate = $1.6 \times 10^{-2}\ M\ s^{-1}$ (A) Calculate the order with respect to each reactant. What is the overall order of the reaction? (2 Marks) (B) Calculate the rate constant and state its units. (2 Marks) (C) Calculate the rate of the reaction when the concentration of both A and B is $0.02\ M$. (1 Mark) [PYQ Term-2 2022] [Confidence: Very High]

Q46. (A) Derive the integrated rate equation for a zero-order reaction. (2 Marks) (B) Identify the order of a reaction from the graph plotting Concentration of R vs. Time yielding a straight line with a negative slope. Write its integrated rate equation mentioning what each term represents. (2 Marks) (C) Show that the half-life of a zero-order reaction is directly proportional to the initial concentration of the reactant. (1 Mark) [PYQ 2025-26 SQP / Practice] [Confidence: High]

Q47. Case Study: Radiocarbon Dating and First-Order Kinetics Radioactive decay follows first-order kinetics. The rate of decay is directly proportional to the number of radioactive nuclei present. The time required for half of the radioactive substance to decay is termed its half-life ($t_{1/2}$), which remains completely independent of the initial concentration. This principle is utilized in Carbon-14 (^{14}C) dating to estimate the age of ancient organic artifacts. (A) What is the overall order of reaction for natural radioactive decay processes? (1 Mark) (B) Write the mathematical formula relating the half-life ($t_{1/2}$) and the decay constant (k). (1 Mark) (C) An archaeological wooden artifact is found to contain only 25% of the ^{14}C activity found in a living tree. Calculate the approximate age of the wooden artifact. ($t_{1/2}$ for $^{14}C = 5730\ years$). (3 Marks) [Practice based on PYQ Themes] [Confidence: Very High]

Q48. (A) Define the terms 'Order of a reaction' and 'Molecularity of a reaction'. Give two points of difference between them. (2 Marks) (B) The rate constant for the first-order decomposition of H_2O_2 is given by the equation: $\log k = 14.2 - \frac{1.0 \times 10^4}{T}$. Calculate E_a for this reaction and the rate constant k if its half-life period is $200\ minutes$. (Given: $R = 8.314\ J\ K^{-1}\ mol^{-1}$). (3 Marks) [PYQ 2016] [Confidence: High]

Q49. (A) According to Collision Theory, what are the two main criteria that must be satisfied for a collision between reactant molecules to yield products? (2 Marks) (B) A first-order reaction is 50% complete in $50\ minutes$ at $300\ K$ and the same reaction is again 50% complete in $25\ minutes$ at $350\ K$. Calculate the activation energy (E_a) of the reaction. (Given: $R = 8.314\ J\ K^{-1}\ mol^{-1}$, $\log 2 = 0.3010$). (3 Marks) [PYQ SQP 2019] [Confidence: Very High]

Q50. (A) Define the term "temperature coefficient" of a chemical reaction. (1 Mark) (B) The rate of a reaction is given by the rate law: $Rate = k[A]^1[B]^2$. How will the rate of the reaction change if the volume of the reaction vessel is suddenly reduced to one-third of its original volume? (2 Marks) (C) Explain why the acid-catalyzed hydrolysis of ethyl acetate is considered a pseudo-first-order reaction even though it is a bimolecular reaction. (2 Marks) [Practice / PYQ Themes] [Confidence: High]

Answer Key

Q1. (B) $-k$ **Q2.** (b) Molecularity has no meaning for complex reactions. **Q3.** (C) $\frac{+1}{3}\frac{d[B]}{dt}$ (Rate = $-\frac{1}{2}\frac{d[A]}{dt} = +\frac{1}{3}\frac{d[B]}{dt}$) **Q4.** (a) $0.00625\ mol\ L^{-1}\ s^{-1}$. ($k = \frac{[A]_0-[A]}{t} = \frac{1.5-0.75}{120} = \frac{0.75}{120} = 0.00625$) **Q5.** (c) increases two times. ($Rate \propto [2A]^2[B/2]^1 = 4 \times 0.5 = 2$) **Q6.** (c) (i) $\ln A$ (ii) $\frac{-E_a}{R}$. (Equation: $\ln k = \ln A - \frac{E_a}{RT}$) **Q7.** (a) ^{15}O. ($t_{1/2} = \frac{0.693}{k}$. Smaller $k \implies$ longer half-life. $5.63 \times 10^{-3} < 2.38 \times 10^{-2}$). **Q8.** (b) Both A and R are true but R is not the correct explanation of A. (Both correctly define rate in terms of reactants and products, but one doesn't explain the other). **Q9.** (c) A is true but R is false. (Order is sum of powers in *rate law*, not stoichiometric coefficients from the balanced equation). **Q10.** (d) A is false but R is true. (Unit of k depends on order: $mol^{1-n}L^{n-1}s^{-1}$. The reason's unit given is for zero order specifically, so reason as a general statement is false too. Wait, if Reason says "The unit of k is moles/L/s" as an absolute, R is false. Answer: Both A and R are false. *Marking point adjustment: If treated strictly, both false.*) **Q11.** (a) Both A and R are true and R is the correct explanation of A. **Q12.** Second order. (Unit formula: $L^{n-1}mol^{1-n}s^{-1} \implies n = 2$). **Q13.** Slope = $-k$. **Q14.** Order = 0, Molecularity = 2. **Q15.** Rate becomes 9 times. ($Rate = k[A]^2 \implies (3)^2 = 9$). **Q16.** $[R] = -kt + [R]_0$. **Q17.** A reaction that appears to be of higher order but follows first-order kinetics because one reactant is in large excess (e.g., acid hydrolysis of ester). **Q18.** A catalyst provides an alternate pathway with a lower activation energy (E_a decreases). **Q19.** A catalyst does not change the Gibbs free energy (ΔG) of the reaction. **Q20.** $k = \frac{0.693}{10} = 0.0693\ min^{-1}$.

Q21. For X: $48\ h = 4$ half-lives. Amount remaining = $1 \times (1/2)^4 = 1/16\ g$. For Y: $48\ h = 3$ half-lives. Amount remaining = $1 \times (1/2)^3 = 1/8\ g$. Ratio X:Y = $1/16 : 1/8 = 1 : 2$. **Q22.** Order = $1/2 + 0 = 1/2$. Unit = $mol^{1-1/2}L^{1/2-1}s^{-1} = mol^{1/2}L^{-1/2}s^{-1}$. **Q23.** (a) Rate decreases (fewer collisions). (b) Rate increases (more molecules cross E_a barrier). **Q24.** Rate of reaction = $\frac{1}{2}\frac{d[C]}{dt} = 1.25 \times 10^{-4}\ M\ s^{-1}$. Rate of disappearance of B = $-\frac{d[B]}{dt} = 3 \times \text{Rate} = 3.75 \times 10^{-4}\ M\ s^{-1}$. **Q25.** Rate of rxn = $\frac{1}{4}\frac{d[NO_2]}{dt} = \frac{2.8\times10^{-3}}{4} = 0.7 \times 10^{-3}$. Disappearance of $N_2O_5 = 2 \times 0.7 \times 10^{-3} = 1.4 \times 10^{-3}\ M\ s^{-1}$. **Q26.** Zero order: $mol\ L^{-1}\ s^{-1}$. Second order: $L\ mol^{-1}\ s^{-1}$. **Q27.** (A) Rate = $k[H_2O_2][I^-]$ (from slow step). (B) Overall order = $1 + 1 = 2$. **Q28.** Intercept = $\ln A$. Slope = $\frac{-E_a}{R} \implies E_a = -Slope \times R$. **Q29.** $k = \frac{0.693}{20} = 0.03465\ min^{-1}$. $t = \frac{2.303}{k}\log(\frac{100}{10}) = \frac{2.303}{0.03465} \times 1 \approx 66.46\ min$. **Q30.** For zero order, $t = \frac{[R]_0-[R]}{k} = \frac{0.1-0.064}{4\times10^{-3}} = \frac{0.036}{0.004} = 9\ seconds$. **Q31.** Time taken for reactant concentration to drop to half its initial value. Zero order: $t_{1/2} = \frac{[R]_0}{2k}$. First order: $t_{1/2} = \frac{0.693}{k}$. **Q32.** Decreasing temp decreases kinetic energy; fewer molecules have energy $\geq E_a$, leading to fewer effective collisions and a decreased rate of reaction.

Q33. $k = \frac{2.303}{80}\log(\frac{100}{60}) = 0.0287 \times 0.2218 \approx 0.00638\ min^{-1}$. For 90%: $t = \frac{2.303}{0.00638}\log(\frac{100}{10}) \approx 361\ min$. **Q34.** $\log(\frac{k_2}{k_1}) = \frac{E_a}{2.303R}(\frac{T_2-T_1}{T_1T_2}) \implies \log 3 = \frac{E_a}{19.147}(\frac{20}{298\times318}) \implies E_a \approx 42.8\ kJ/mol$. **Q35.** $\log(\frac{0.20}{0.02}) = \log 10 = 1.1 = \frac{E_a}{19.15} \times (\frac{300}{100000}) \implies E_a \approx 6383\ J/mol$ or $6.38\ kJ/mol$.

Q36. At 300K, $k_1 = \frac{0.693}{30} = 0.0231$. At 320K, $k_2 = \frac{0.693}{100} = 0.00693$. (Wait, rate usually increases with temp. Let's assume PYQ meant inverse or negative Ea, but applying formula gives $E_a = \frac{2.303R \cdot T_1 T_2}{T_2 - T_1} \log(\frac{k_2}{k_1})$. Since $k_2 < k_1$, E_a is negative, which is an anomaly or catalytic case). **Q37.** $k_1 = 0.693/40$, $k_2 = 0.693/20 \implies k_2/k_1 = 2$. $\log 2 = \frac{E_a}{19.147}(\frac{20}{300 \times 320}) \implies E_a \approx 27.66\ kJ/mol$. **Q38.** $k_2/k_1 = 4$. $\log 4 = \frac{E_a}{19.147}(\frac{20}{293 \times 313}) \implies 0.6021 = \frac{E_a \times 20}{19.147 \times 91709} \implies E_a \approx 52.8\ kJ/mol$. **Q39.** $k_2/k_1 = 2$. $\log 2 = \frac{E_a}{19.147}(\frac{10}{300 \times 310}) \implies 0.301 = \frac{E_a \times 10}{19.147 \times 93000} \implies E_a \approx 53.6\ kJ/mol$. **Q40.** Order = 0. Molecularity = 2 (bimolecular). Unit of k = $mol\ L^{-1}\ s^{-1}$. At high pressure, metal surface gets fully saturated, so rate becomes independent of further gas concentration (zero order). **Q41.** Comparing to $k = Ae^{-E_a/RT} \implies \frac{E_a}{R} = 25000 \implies E_a = 25000 \times 8.314 = 207.85\ kJ/mol$. If $t_{1/2} = 300\ min = 18000\ s$, $k = \frac{0.693}{18000} = 3.85 \times 10^{-5}\ s^{-1}$. **Q42.** $t_{99\%} = \frac{2.303}{k}\log(\frac{100}{1}) = \frac{2.303}{k} \times 2$. $t_{90\%} = \frac{2.303}{k}\log(\frac{100}{10}) = \frac{2.303}{k} \times 1$. Thus, $t_{99\%} = 2 \times t_{90\%}$.

Q43. (A) $\Delta H = E_{products} - E_{reactants} = 100 - 110 = -10\ kJ/mol$ (Exothermic). (B) Backward $E_a = 140 - 100 = 40\ kJ/mol$. (C) A positive catalyst lowers the activation energy, making the peak lower, which speeds up both forward and backward rates. (D) No, a catalyst does not change the energy of reactants or products, so equilibrium constant K remains unaffected. **Q44.** (A) From Rate = $-\frac{d[R]}{dt} = k[R]$, separating variables and integrating gives $[R] = [R]_0 e^{-kt}$, turning to log gives $k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$. (B) $t = \frac{2.303}{1.15 \times 10^{-3}}\log(\frac{5}{3}) = 2002.6 \times (0.6990 - 0.4771) = 2002.6 \times 0.2219 \approx 444\ seconds$. **Q45.** (A) Doubling [A] (Exp 1 to 2) doubles rate $\implies$ Order w.r.t A = 1. Doubling [B] (Exp 2 to 3) quadruples rate ($4 \to 16$) $\implies$ Order w.r.t B = 2. Overall = 3. (B) $2.0 \times 10^{-3} = k(0.1)^1(0.1)^2 \implies k = 2.0\ L^2\ mol^{-2}\ s^{-1}$. (C) Rate = $2.0 \times (0.02)^1 \times (0.02)^2 = 1.6 \times 10^{-5}\ M\ s^{-1}$. **Q46.** (A) Rate = $-\frac{d[R]}{dt} = k \implies d[R] = -kdt$. Integrating $\implies [R] = -kt + C$. At $t = 0, [R] = [R]_0 \implies C = [R]_0$. Equation: $[R] = -kt + [R]_0$. (B) The graph corresponds to a zero-order reaction. Integrated rate eq: $k = \frac{[R]_0 - [R]}{t}$. (C) At $t = t_{1/2}$, $[R] = [R]_0/2$. So $t_{1/2} = \frac{[R]_0 - [R]_0/2}{k} = \frac{[R]_0}{2k} \implies t_{1/2} \propto [R]_0$. **Q47.** (A) First Order. (B) $k = \frac{0.693}{t_{1/2}}$. (C) Activity drops to 25%, meaning it has undergone 2 half-lives ($100\% \to 50\% \to 25\%$). Age = $2 \times 5730 = 11,460\ years$. **Q48.** (A) $k = Ae^{-E_a/RT}$. A = Arrhenius factor, E_a = Activation energy, R = Gas constant, T = Temperature in Kelvin. (B) Comparing to $\log k = \log A - \frac{E_a}{2.303RT}$, $\frac{E_a}{2.303R} = 1.0 \times 10^4 \implies E_a = 10000 \times 2.303 \times 8.314 = 191.47\ kJ/mol$. $k = \frac{0.693}{200} = 3.465 \times 10^{-3}\ min^{-1}$. **Q49.** (A) 1. Proper orientation of molecules. 2. Molecules must possess minimum threshold energy (Activation Energy). (B) $k_1 = \frac{0.693}{50}$, $k_2 = \frac{0.693}{25} \implies k_2/k_1 = 2$. $\log 2 = \frac{E_a}{19.147}(\frac{350 - 300}{300 \times 350}) \implies 0.3010 = \frac{E_a \times 50}{19.147 \times 105000} \implies E_a \approx 12.08\ kJ/mol$. **Q50.** (A) The ratio of the rate constant of a reaction at two temperatures differing by $10^\circ C$ (usually 298 K and 308 K). (B) Volume reduced to 1/3 means concentration increases by 3 times for all reactants. $Rate_{new} = k[3A]^1[3B]^2 = 27 \times k[A][B]^2$. Rate becomes 27 times. (C) Ethyl acetate hydrolysis uses water as a reactant. Since water is the solvent and present in huge excess, its concentration hardly changes. Thus, the rate law practically depends only on the ester, making it pseudo-first-order.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).